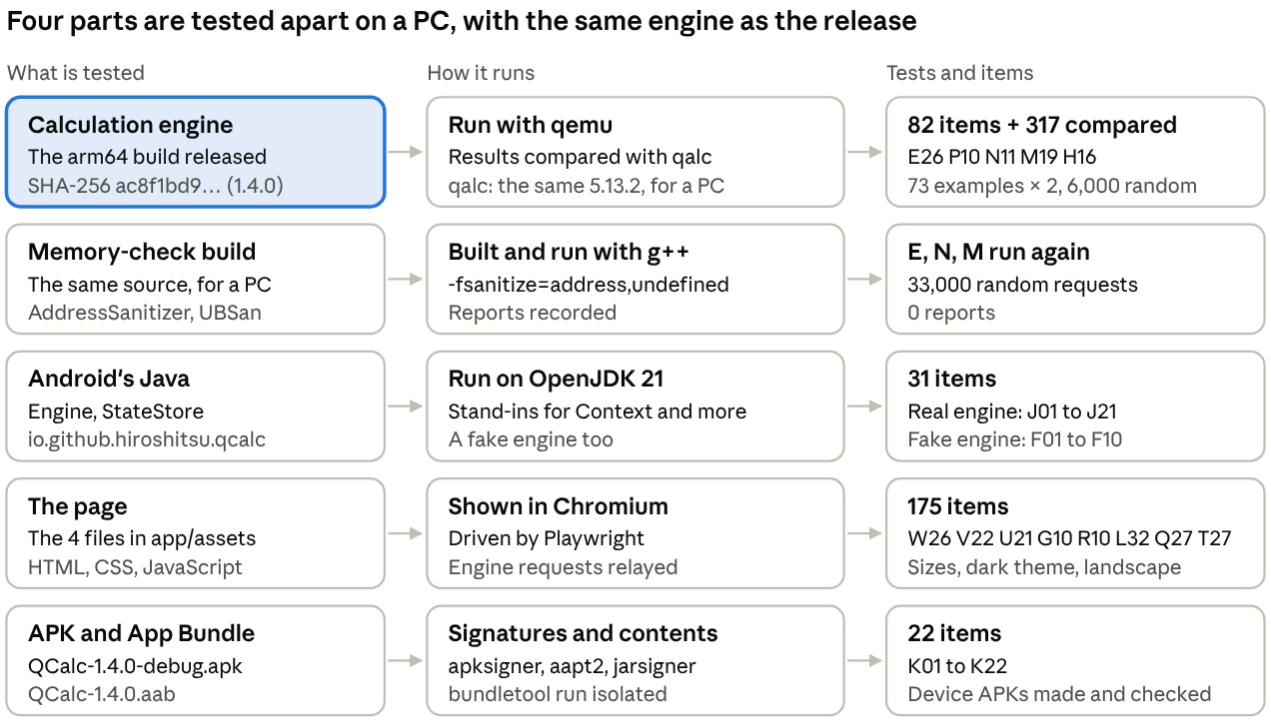
6. Test specification and results

QCalc 1.4.0 passed all the tests in a simulated environment on a PC (run on 2026-11-13). It has not been tested on a real phone yet.

6.1 Test approach and environment

As no phone was at hand, the four parts (the calculation engine, the Android Java code, the page, and the APK and App Bundle) were checked separately on a PC. Results are checked against the official qalc, built for a PC from the same source. Version 1.1.0 added tests of calculus (derivatives, integrals, limits, differential equations) and matrices, the Japanese units, the tabs of keys, the examples, Undo and text sizes, and raised the comparison with the official qalc from 30 to 309 expressions. The engine's source was also built for a PC with memory checks (AddressSanitizer and UndefinedBehaviorSanitizer), and the same tests and random requests showed no memory errors or undefined behavior. Version 1.2.0, which raised the precision limit to 200000 digits, added tests of high precision (H01 to H16), of carrying ans over with all its digits (J15 to J21) and of long results on the page (L01 to L32), checking every digit of the results against another library (Python's mpmath). Version 1.3.0 added tests of the English and Japanese screens, QCalc Pro, color themes, exporting the history (CSV), all digits in the preview and digit grouping (Q01 to Q27). The comparison with the official qalc (309 expressions), the engine's function tests, 3D graphs, calculus and matrices, the Japanese units, the examples (of both the Japanese and the English screen), the 31 Java tests, the tests of the Japanese screen (tabs of keys; examples, Undo, text size and search; 3D graphs; fixes from the review) and the 22 checks of the APK and App Bundle were run again with 1.3.0 (2026-11-13). The tests of high precision (H), long results on the page (L) and reusing the history, input and operation (U) were not run again then and are results of version 1.2.0; what changed in 1.3.0 is noted under each table. Version 1.4.0 added tests of the suggestions for names and conversion targets, the lines of the history, the Light or dark setting, the landscape layout and the wrapping of long numbers (T01 to T27), and 8 expressions with “to mixed” to the comparison with the official qalc. All the tests except those of high precision (H) were run again with 1.4.0 (2026-11-13), including long results on the page (L) and reusing the history, input and operation (U), on the Japanese screen. The tests of high precision (H) are results of version 1.2.0.



How the tests are organized · four parts and the test tools

Item	Contents
OS	Ubuntu 24.04.5 LTS (x86-64)
Calculation engine	The same engine as in the APK and App Bundle (1.4.0: SHA-256 ac8f1bd9…b9a7, 7,588,488 bytes; 1.3.0: 1b57c372…102e, 7,588,488 bytes; 1.2.0: 56576cc9…1a50, 7,587,080 bytes; the tests of 1.1.0: 09ec8685…ce07a, 7,582,440 bytes), run with qemu-aarch64-static (8.2.2)
The official qalc for comparison	qalc built for a PC from libqalculate's tag v5.13.2 (commit d3672af9), calculating with qalc -defaults (the default settings)
Memory checks	qalc-engine.cc linked with libqalculate 5.13.2 built for a PC, and built with g++ -fsanitize=address,undefined (AddressSanitizer and UndefinedBehaviorSanitizer). The reports (standard error) are written to a file

Item	Contents
Java	Engine.java and StateStore.java (package io.github.hiroshitsu.qcalc) run as they are on OpenJDK 21.0.12. Android's Context and SharedPreferences are small stand-ins for the tests; org.json is Ubuntu's com.android.json
The page	app/assets shown in Chromium (Playwright), with the requests relayed to the engine above. Screen size 412 × 854 (close to the AQUOS sense8; 405 × 720 for the 3D graph tests; the tabs of keys were also checked at 360 × 740, 320 × 568 and landscape 740 × 360; Q01 to Q27 of 1.3.0 at 412 × 915, on the English and the Japanese screen)
APK and App Bundle	apksigner 31.0.2, zipalign 10.0.0, aapt2 2.19, jarsigner (OpenJDK 21.0.12), bundletool 1.17.2 (downloaded, so run as a dedicated user cut off from the network)

Summary of the results

Group	Items	Passed
Comparison with the official qalc (6.2; run again with 1.4.0)	317	317
Engine function tests (6.2; run again with 1.4.0)	26	26
3D graph calculation (plot3d) (6.2; run again with 1.4.0)	10	10
Calculus and matrices (6.2; run again with 1.4.0)	19	19
Japanese units (6.2; run again with 1.4.0)	11	11
All the examples calculated (6.2; the expressions of the Japanese and the English screen of 1.4.0)	73	73
Java with the real engine (6.3; run again with 1.4.0)	14	14
Java with a fake engine (6.3; run again with 1.4.0)	10	10
Tabs of keys, calculus and matrices on the page (6.4; run again with 1.4.0)	26	26

Group	Items	Passed
Examples, Undo, text size, search (6.4; run again with 1.4.0)	22	22
Reusing the history, converting input, operation (6.4; run again with 1.4.0)	21	21
3D graphs on the page (6.4; run again with 1.4.0)	10	10
Fixes from the review (6.4; run again with 1.4.0)	10	10
APK and App Bundle checks (6.5; with the files of 1.4.0)	22	22
High precision (6.2; version 1.2.0)	16	16
Java: ans with all its digits (6.3; run again with 1.4.0)	7	7
Long results on the page (6.4; version 1.2.0; run again with 1.4.0)	32	32
The page of 1.3.0 (QCalc Pro, color themes, export, preview, digit grouping) (6.4; run again with 1.4.0)	19	19
Language of the screen (6.4; version 1.3.0; run again with 1.4.0)	7	7
No Japanese left on the English screen (6.4; version 1.3.0; run again with 1.4.0)	1	1
The page of 1.4.0 (suggestions, the history, Light or dark, landscape, wrapping) (6.4)	27	27
Total	700	700

In addition, the build with memory checks ran the function tests and 33,000 random requests, and the released engine 6,000 random requests, with no problems (end of 6.2). In 1.2.0, the comparison with the official qalc and the tests of long results on the page were also run with the build with memory checks, and no errors were reported. In 1.4.0, a test that taps keys, tabs, menu items and history lines at random (on a 360 × 640 screen, 350 actions with each of 3 random seeds) was also run, with no script errors on the page.

6.2 Tests of the calculation engine and results

Comparison with the official qalc (all 317 the same)

The same expressions were calculated with the official qalc and with the app's engine (default settings), and the result strings were checked to be the same. qalc shows "expression = result", so its result part is compared. Version 1.1.0 raised the number to 309 expressions, and version 1.4.0 to 317 by adding 8 expressions with "to mixed" (breakdown below). Below are the 30 basic expressions kept since 1.0.0 and the 56 expressions of calculus and matrices added in 1.1.0. The expression with a Japanese unit (1 tsubo to m^2) is left out of the comparison, as the official qalc does not have the unit, and is checked by the tests of the Japanese units (N01 to N11). In 1.2.0 and 1.3.0 the same 309 expressions were compared again, and all were the same (the engine of 1.3.0 took 16 seconds in qemu). In 1.4.0 the 317 expressions were compared, and all were the same (17 seconds in qemu).

No	Expression	Result (the same in the official qalc and the app's engine)
C-01	1+2*3	7
C-02	1/3 + 1/4	7/12 ≈ 0.5833333333
C-03	0.1 + 0.2	0.3
C-04	sqrt(8)	2 × √(2) ≈ 2.828427125
C-05	sin(30 deg)	1/2 = 0.5
C-06	2^100	1.267650600E30
C-07	100!	9.332621544E157
C-08	5 km to mi	3 mi + 188 yd + 2.393700787 in
C-09	60 km/h * 2.5 h	150 km
C-10	20 °C to °F	68 °F
C-11	solve(x^2 - 5x + 6)	[3 2]
C-12	x^2 = 4	x = 2 or x = −2
C-13	integrate(x^2 sin(x), 0, pi)	π² − 4 ≈ 5.869604401
C-14	diff(x^3)	3x²
C-15	sum(n^2, 1, 10)	385
C-16	[[1,2],[3,4]]^-1	[−2 1; 3/2 −1/2] = [−2 1; 1.5 −0.5]
C-17	255 to hex	0xFF
C-18	0b1010 + 0x1F	41

No	Expression	Result (the same in the official qalc and the app's engine)
C-19	factor 360	$2^3 \times 3^2 \times 5$
C-20	gcd(12, 18)	6
C-21	2000 * 10%	200
C-22	10 mod 3	1
C-23	(1 + 2i)^2	−3 + 4i
C-24	e^(i*pi)	−1
C-25	(5 +/- 0.2) m * (3 +/- 0.1) m	15.00±0.78 m²
C-26	1 GiB to MB	1073.741824 MB
C-27	speed_of_light to km/s	299792.458 km/s
C-28	"2026-11-09" + 100 days	"2027-02-17"
C-29	7/3	2 + 1/3 ≈ 2.333333333
C-30	cos(pi/3) + ln(e^2)	5/2 = 2 + 1/2 = 2.5

Breakdown of the expressions compared

Set	Expressions	Main contents
C-01 to C-30 (the 30 basic ones)	30	Arithmetic, roots, trigonometric functions, units, equations, integrals, derivatives, sums, inverse matrices, number bases, factorization, percentages, remainders, complex numbers, uncertainty, physical constants, dates

Set	Expressions	Main contents
C-31 to C-253 (a wide range)	223	Functions (factorials, combinations, rounding, exponentials, logarithms, trigonometric and hyperbolic functions), complex numbers, unit conversion, equations, inequalities and systems of equations, derivatives, integrals, limits and sums, basic matrices and vectors, primes and factorization, bit operations, binary, octal and hexadecimal numbers and Roman numerals, fractions, dates and times, uncertainty, physical constants, statistics, special functions, strings, symbolic expressions (factorization, expansion), and where an exact form is shown before an approximation (long exact forms)
C-254 to C-309 (calculus and matrices)	56	Derivatives 7, extrema 1, integrals 10, numerical integration 1, limits 1, sums and products 3, differential equations 3, numerical solutions and critical points 2, special functions 3, complex numbers 1, matrices and vectors 24
C-310 to C-317 (mixed units; 1.4.0)	8	“to mixed” (lengths, masses, times), checking that the added Japanese units do not appear in the results

Comparison of calculus and matrices (the 56 expressions added in 1.1.0)

No	Expression	Result (the same in the official qalc and the app's engine)
C-254	diff(x^3 + 2x)	$3x^2 + 2$
C-255	diff(x^2 × sin(x))	$\cos(x) \times x^2 + 2 \times \sin(x) \times x$
C-256	diff(x^5, x, 2)	$20x^3$
C-257	diff(x^2 × y^3, y)	$3x^2y^2$

No	Expression	Result (the same in the official qalc and the app's engine)
C-258	diff(x^3, x, 1, 2)	12
C-259	diff(ln(x^2 + 1))	(2x) / (x² + 1)
C-260	diff(sin(x)^2, x, 3)	−4 × sin(2x)
C-261	extremum(x^3 − 3x)	[1 −1]
C-262	integrate(1/x)	ln(|x|) + C
C-263	integrate(x^2 × sin(x), 0, π)	π² − 4 ≈ 5.869604401
C-264	integrate(√(1 − x^2), −1, 1)	π / 2 ≈ 1.570796327
C-265	integrate(e^(−x^2), −∞, ∞)	√(π) ≈ 1.772453851
C-266	integrate(1/x^2, 1, ∞)	1
C-267	integrate(1/(1 + x^2), 0, ∞)	π / 2 ≈ 1.570796327
C-268	integrate(integrate(x × y, 0, 1, x), 0, 2, y)	1
C-269	integrate(e^(−x^2), 0, 1)	(√(π) × erf(1)) / 2 ≈ 0.7468241328
C-270	integrate(x × ln(x), 1, e)	(e² + 1) / 4 ≈ 2.097264025
C-271	integrate(tan(x))	−ln(|cos(x)|) + C
C-272	romberg(sin(x)/x, 1, 2)	0.6593299064
C-273	limit((x^2 − 1)/(x − 1), 1)	2
C-274	sum(x^2, 1, 10)	385

No	Expression	Result (the same in the official qalc and the app's engine)
C-275	sum(1/2^n, 0, 20, n)	1.999999046
C-276	product(x, 1, 5)	120
C-277	dsolve(diff(y, x) = y)	e^x × C ≈ 2.718281828^x × C
C-278	dsolve(diff(y, x) = −2y, 5)	5 / e^(2x) ≈ 5.000000000 × 0.1353352832^x
C-279	dsolve(diff(y, x) + 2y = e^x, 1)	e^x / 3 + 2 / (3e^(2x)) ≈ 0.3333333333 × 2.718281828^x + 0.6666666667 × 0.1353352832^x
C-280	newtonsolve(cos(x) = x, 1)	0.7390851332
C-281	solve(diff(x^3 − 3x) = 0)	[1 −1]
C-282	gamma(1/2)	√(π) ≈ 1.772453851
C-283	zeta(2)	π² / 6 ≈ 1.644934067
C-284	Si(1)	0.9460830704
C-285	[1, 2; 3, 4]	[1 2; 3 4]
C-286	[1 2; 3 4]	[1 2; 3 4]
C-287	[1, 2; 3, 4] × [5, 6; 7, 8]	[19 22; 43 50]
C-288	[1, 2; 3, 4]^−1	[−2 1; 3/2 −1/2] = [−2 1; 1.5 −0.5]
C-289	[1, 2; 3, 4]^2	[7 10; 15 22]
C-290	2 × [1, 2; 3, 4] + 1	[3 5; 7 9]

No	Expression	Result (the same in the official qalc and the app's engine)
C-291	det([1, 2, 3; 4, 5, 6; 7, 8, 10])	−3
C-292	transpose([1, 2, 3; 4, 5, 6])	[1 4; 2 5; 3 6]
C-293	adj([1, 2; 3, 4])	[4 −2; −3 1]
C-294	rk([1, 2; 2, 4])	1
C-295	rref([2, 1, 3; 1, 3, 5])	[1 0 4/5; 0 1 7/5] = [1 0 0.8; 0 1 1.4]
C-296	identity(3)	[1 0 0; 0 1 0; 0 0 1]
C-297	[2, 1; 1, 3]^−1 × [3; 5]	[4/5; 7/5] = [0.8; 1.4]
C-298	multisolve([2x + y = 3, x + 3y = 5], [x, y])	[4/5 7/5] = [0.8 1.4]
C-299	solve(det([2, 1; 1, 2] − x × identity(2)) = 0)	[3 1]
C-300	[1, 2; 3, 4] × [x; y]	[(x + 2y); (3x + 4y)]
C-301	[[1, 2, 3], [4, 5, 6], [7, 8, 10]]^−1	[−2/3 −4/3 1; −2/3 11/3 −2; 1 −2 1] ≈ [−0.6666666667 −1.333333333 1; −0.6666666667 3.666666667 −2; 1 −2 1]
C-302	dot([1, 2, 3], [4, 5, 6])	32
C-303	norm([3, 4])	5
C-304	[1, 2, 3] + 2 × [4, 5, 6]	[9 12 15]
C-305	[1, 2, 3] × [4, 5, 6]	[1 2 3] × [4 5 6]
C-306	[1, 2; 2, 4]^−1	1 / [1 2; 2 4]

No	Expression	Result (the same in the official qalc and the app's engine)
C-307	(3 + 4i) × (1 − 2i)	11 − 2i
C-308	kron([1, 2; 3, 4], [1, 0; 0, 1])	[1 0 2 0; 0 1 0 2; 3 0 4 0; 0 3 0 4]
C-309	exp([1, 2; 3, 4])	[e e^2; e^3 e^4] ≈ [2.718281828 7.389056099; 20.08553692 54.59815003]

C-258's diff(x^3, x, 1, 2) is the first derivative at x = 2. C-299 finds the eigenvalues of the matrix [2, 1; 1, 2] (3 and 1) from the characteristic equation. C-305 and C-306 cannot be calculated, and return the expression as it is with an error (use cross(), dot() or hadamard() for a product of vectors; the matrix has no inverse), as qalc does.

Comparison of mixed units (the 8 expressions added in 1.4.0)

“to mixed” writes a result with larger and smaller units together. Up to 1.3.0 the added Japanese units were chosen as units to mix, so that C-310 gave “16.5 寸 + 1 m” and C-311 “1 貫 + 1.25 kg”, unlike the official qalc (detailed design 3.19). 1.4.0 fixes this, and all 8 expressions give the same results.

No	Expression	Result (the same in the official qalc and the app's engine)
C-310	1.5 m to mixed	1.5 m
C-311	5 kg to mixed	10 pfund
C-312	3.7 h to mixed	3 h + 42 min
C-313	10000 s to mixed	2 h + 46 min + 40 s
C-314	100 in to mixed	8 ft + 4 in
C-315	5.5 ft to mixed	5 ft + 6 in
C-316	2.5 lb to mixed	2 lb + 8 oz
C-317	1.25 mi to mixed	1 mi + 440 yd

Function tests (all 26 passed; run again with the engine of 1.4.0)

One engine was started and given requests in turn, and the replies were checked (QALC_CURRENCY=JPY).

ID	What	Checked (actual values)	Result
E01	Reply at start	engine is libqalculate 5.13.2; in a new folder, ans_defined=false	Pass
E03	The preview does not run assignments	x = 5 gives deferred=true, and the following x + 1 stays x + 1	Pass
E04	The preview does not run save()	deferred=true	Pass
E05a	The ans event after committing	{"event":"ans","id":5,"value":"7/3","input":"7/3"}	Pass
E05b	ans equals the value shown	ans × 3 = 7	Pass
E06	The solutions of an equation go into ans	the input of x = 2 or x = −2 is [2 −2]	Pass
E07	ans2 holds the previous result	ans2 = 7/3	Pass
E08	set_ans brings ans back	set=true, ans + 1 = 11	Pass
E09	max_chars cuts long results	100! (all digits) cut at 100 bytes with "…", truncated=true	Pass
E10	Broken JSON refused	{"id":null,"ok":false,"error":"bad request"}, the same with extra characters after it	Pass
E11	Too deeply nested expressions refused	301 levels of brackets give "The expression is nested too deeply" (the engine's texts are in English since 1.3.0; the Japanese screen shows 式の入れ子が深すぎます)	Pass
E12	Unknown command	unknown command	Pass
E13a	Defining and using a variable	r = 5 cm saved, r × 2 = 100 mm	Pass
E13b	Defining and using a function	hyp = sqrt(x^2 + y^2) saved, hyp(3, 4) = 5	Pass
E13c	What cannot be defined is refused	pi and sin are built-in names, 1abc has characters that cannot be used, 5 +* 3 gives "Check the value …" (in English since 1.3.0)	Pass
E14	Definitions saved to files	definitions/variables.xml and functions.xml are made, and user_defs lists r and hyp	Pass

ID	What	Checked (actual values)	Result
E15	Deleting a definition	saved=true the first time; the second time “There is no user definition "r"” (in English since 1.3.0)	Pass
E16	A committed x = 5 is a temporary variable	temporary=true in user_defs	Pass
E17	Graph points	2 series, sin(x) and 1/x, of 9 points each (steps 10)	Pass
E18	Checks of the graph range and the number of expressions	min > max gives invalid range; 13 expressions give too many expressions	Pass
E19	Lists	426 functions, 192 variables, 438 units (functions: libqalculate's 424, trace() and hyp defined in E13b; units: with the 10 Japanese units)	Pass
E20	Exchange-rate sources	eurofxref-daily.xml, btc.json, rates.json (all in the user's folder)	Pass
E21	The phone's currency	100 USD is shown in ¥	Pass
E22	Timeout and restart	fibonacci(10^9) in exact mode (timeout 1500): timeout=true and restart=true after 4.6 seconds, exit code 3	Pass
E23	Long exact forms are not shown before the approximation (as in qalc)	pi + e + sqrt(2) + ln(2) + sin(1) + cos(1) + tan(1) gives only 10.90641624 (approx=true). fibonacci(10^9) in automatic mode does not time out and answers with an approximation (starting 9.694687234E…)	Pass
E24	The unit list returns text that can be typed for compound units	km, km/h and GB are returned; with them, 5 km + 3 cm to m = 5000.03 m and 90 km/h to m/s = 25 m/s. 11 units cannot be inserted into an expression (insert=false)	Pass

The value of E21 comes from the exchange rates built into the engine (they cannot be fetched in the test environment).

3D graph calculation (plot3d) (all 10 passed; run again with 1.4.0)

plot3d requests were sent to the engine, and the grids and z values returned were checked.

ID	What	Checked (actual values)	Result
P01	Shape and values of the grid	sin(x) cos(y), 5 × 5: x is −5, −2.5, 0, 2.5, 5; z has 25 values, the first sin(−5)·cos(−5) = 0.2720105554	Pass
P02	Two expressions, and points that are not real	x*y and sqrt(x) (−1 to 1, 3 × 3): z is [1, 0, −1, 0, 0, 0, −1, 0, 1] and [null, 0, 1, null, 0, 1, null, 0, 1]	Pass
P03	Poles (points without a value)	1/(x*y) (5 × 5): the 9 points where x = 0 or y = 0 (counted from 0: 2, 7, 10 to 14, 17, 22) are null	Pass
P04	Values with units are not drawn	x * 1 m: all 9 points null	Pass
P05	Ranges can be expressions	x from pi to 2pi (3.14159265359, 4.71238898038, 6.28318530718), y from 0 to 1	Pass
P06	Wrong ranges refused	x from 5 to 1 gives invalid x range; y = abc gives invalid y range	Pass
P07	Number of expressions checked	5 give too many expressions, 0 give exprs missing	Pass
P08	Limits of the grid	nx 500 becomes 121, ny 1 becomes 2	Pass
P09	Time for a fine grid	61 × 61 (3,721 points) in 0.2 seconds with qemu	Pass
P10	Timeout	fibonacci(200000 + round(abs(x))) (41 × 41, timeout 1500): complete=false after 1.5 seconds, without restarting the engine	Pass

Calculus and matrices (all 19 passed; run again with 1.4.0)

The expressions typed by the keys added in 1.1.0 (Calculus, Matrix), trace() added by the bridge program, and the fix for the message hold problem were checked. The expressions are written with the characters the keys type (×, −, π, ∞ and so on).

ID	What	Checked (actual values)	Result
M01	Derivatives (first, second, partial, at a point) and extrema	diff(x^3 + 2x) = 3x² + 2, diff(x^5, x, 2) = 20x³, diff(x^2 × y^3, y) = 3x²y², diff(x^3, x, 1, 2) = 12, diff(x^2 × sin(x)) = cos(x) × x² + 2 × sin(x) × x, extremum(x^3 − 3x) = [1 −1]	Pass
M02	Integrals (indefinite, definite, double)	integrate(1/x) = ln(|x|) + C, integrate(x^2 × sin(x), 0, π) = π² − 4 ≈ 5.869604401, integrate(√(1 − x^2), −1, 1) = π / 2, integrate(integrate(x × y, 0, 1, x), 0, 2, y) = 1 (the inner one alone gives y / 2)	Pass
M03	Integrals with an infinite bound	integrate(e^(−x^2), −∞, ∞) = √(π) ≈ 1.772453851, integrate(1/x^2, 1, ∞) = 1, integrate(1/(1 + x^2), 0, ∞) = π / 2, integrate(e^(−x), 0, ∞) = 1	Pass
M04	Warnings still appear after M03 (the fix for the libqalculate 5.13.2 problem)	“Division by zero.” for 1/0; limit(sin(x)/x, 0) = 1 with “limit() is incomplete and unreliable.”; a warning also for an expression with an integral and 1/0 (= 1 + 1 / 0), and for the following 1/0	Pass
M05	After the same integral 25 times (committed and previewed in turn)	integrate(e^(−x), 0, ∞) + 5 = 6 without messages; a warning for 1/0 both committed and previewed	Pass
M06	Limits (at a point, at infinity, from the right, from the left)	limit(sin(x)/x, 0) = 1, limit((1 + 1/x)^x, ∞) = e, limit(1/x, 0, x, 1) = +∞, limit(1/x, 0, x, −1) = −∞, limit((x^2 − 1)/(x − 1), 1) = 2	Pass
M07	Sums and products	sum(x^2, 1, 10) = 385, product(x, 1, 5) = 120, sum(1/2^n, 0, 20, n) = 1.999999046, sum(x, 1, 100) = 5050	Pass
M08	Differential equations, numerical solutions, critical points	dsolve(diff(y, x) = y) = e^x × C, dsolve(diff(y, x) = −2y, 5) = 5 / e^(2x), dsolve(diff(y, x) = x^2) = x³ / 3 + C, newtonsolve(cos(x) = x, 1) = 0.7390851332, solve(diff(x^3 − 3x) = 0) = [1 −1]	Pass

ID	What	Checked (actual values)	Result
M09	Special functions	gamma(1/2) = √(π), zeta(2) = π² / 6, erf(1) = 0.8427007929, Si(1) = 0.9460830704, integrate(e^(−x^2), 0, 1) = (√(π) × erf(1)) / 2	Pass
M10	Three ways to write a matrix; product, inverse and power	[1, 2; 3, 4], [[1, 2], [3, 4]] and [1 2; 3 4] all give [1 2; 3 4]. Product [19 22; 43 50], inverse [−2 1; 3/2 −1/2] = [−2 1; 1.5 −0.5], square [7 10; 15 22], 2 × A + 1 = [3 5; 7 9]	Pass
M11	det, transpose, adj, rk, rref, identity	det = −3, transpose [1 4; 2 5; 3 6], adj [4 −2; −3 1], rk([1, 2; 2, 4]) = 1, rref [1 0 4/5; 0 1 7/5], identity(3), A^{-1} × A = [1 0; 0 1]	Pass
M12	Systems of equations (two ways), eigenvalues, matrices with symbols	[2, 1; 1, 3]^−1 × [3; 5] = [4/5; 7/5], [4/5 7/5] with multisolve, solve(det([2, 1; 1, 2] − x × identity(2)) = 0) = [3 1], [1, 2; 3, 4] × [x; y] = [(x + 2y); (3x + 4y)]	Pass
M13	Vectors, and errors for what cannot be calculated	dot = 32, cross = [0 0 1], norm([3, 4]) = 5, [1, 2, 3] + 2 × [4, 5, 6] = [9 12 15]. vector × vector gives an error asking for cross(), dot() and the like; [1, 2; 2, 4]^−1 gives “Inverse of singular matrix.”; a product of the wrong sizes is an error too	Pass
M14	trace() (added by the bridge program)	trace([1, 2; 3, 4]) = 5, trace([1/2, x; y, 1/3]) = 5/6, trace([1 m, 0; 0, 2 m]) = 3 m, trace(identity(4)) = 4, trace([1, 2, 3; 4, 5, 6; 7, 8, 10]) + det([1, 2; 3, 4]) = 14. A matrix that is not square gives the error “must be a square matrix.”	Pass
M15	How trace() is registered	1 item in Matrices & Vectors of the function list (title Trace; not a user definition), 425 functions in all. No user function named trace can be made. A user function tr2 that uses trace can be made: tr2([4, 0; 0, 6]) = 5. trace is not written to the definitions file	Pass

ID	What	Checked (actual values)	Result
M16	A matrix in ans	after committing [1, 2; 3, 4]^−1, det(ans) = −1/2 and ans × [1, 2; 3, 4] = [1 0; 0 1]. The text to insert from the ans event, [−2 1; 1.5 −0.5], gives the same matrix when calculated again	Pass
M17	Graphs of a derivative and of integrals with a variable upper bound	3 lines: diff(x^3 − 3x), integrate(2y, 0, x, y) and integrate(e^(−y^2), 0, x, y). At x = −2, −1, 0, 1, 2: 9, 0, −3, 0, 9; 4, 1, 0, 1, 4; and −0.882081, −0.746824, 0, 0.746824, 0.882081	Pass
M18	Matrices with each display setting (shown as in qalc)	Exact: [−2/3 1/3; 1/2 −1/6] = [−2/3 1/3; 0.5 −1/6]. Approximate and automatic: … ≈ [−0.6666666667 0.3333333333; 0.5 −0.1666666667]. Hexadecimal: [0xA 0xB; 0xC 0xD]	Pass
M19	After the engine restarts	trace([1, 2; 3, 4]) = 5, the user function tr2([4, 0; 0, 6]) = 5	Pass

Japanese units (all 11 passed; run again with 1.4.0)

The 10 units added by the bridge program (detailed design 3.19) were checked.

ID	What	Checked (actual values)	Result
N01	Tsubo and square metres	30 坪 to m² = 99.17355372 m², 100 m² to 坪 = 30.25 坪	Pass
N02	The Latin name tsubo; 1 tsubo = 36 square shaku	1 tsubo to m^2 = 3.305785124 m², 1 坪 to 尺^2 = 36 尺²	Pass
N03	Lengths: shaku, ken, sun	6 尺 = 181.8181818 cm, 1 間 = 6 尺, 10 寸 = 1 尺	Pass
N04	Ri	1 里 = 3.927272727 km, 2 km = 0.5092592593 里	Pass
N05	Volumes: gō, shō	3 合 = 541.1720511 mL, 1 升 = 10 合, 1.8 L = 0.9978342357 升	Pass
N06	Masses: kan, monme	1 貫 = 3.75 kg, 8 匁 = 30 g, 60 kg = 16 貫	Pass
N07	Tan = 300 tsubo	2 反 = 600 坪, 1 反 = 9.917355372 a	Pass

ID	What	Checked (actual values)	Result
N08	Calculating with the units	3 坪 + 2 m^2 = 11.91735537 m², 80 m² / 坪 = 121/5 = 24.2, 25 坪 at ¥350,000 per tsubo = ¥8750000	Pass
N09	Listed in their own category (Japanese Units), not as user definitions	坪, 反, 寸, 尺, 間, 里, 合, 升, 匁, 貫	Pass
N10	Protected names; a user variable after a restart	No variable named 坪 can be made (“坪” is a built-in name). madori = 6 坪 is still 19.83471074 m² after a restart. The units are not written to the user’s files	Pass
N11	Short Latin names are not read as Japanese units	tan(45°) + 1 = 2 (tan stays the tangent). go and sun are not read as 合 and 寸 either	Pass

Examples (all 73 calculated; the expressions of the Japanese and the English screen of 1.4.0)

The 69 examples in the app and the 4 on the first screen were sent to the engine with the characters shown on the screen (×, −, √, π and so on). As the app does, each was calculated twice, as a preview (time limit 1.5 seconds) and committed (store), and all 73 met the conditions: both give a result, there is no error or timeout, the preview and the committed result are the same, and the preview does not hold back the calculation (deferred).

In 1.3.0, the expressions of the Japanese and the English screen (73 each) were calculated with the engine of 1.3.0, including the 4 that differ on the English screen (100 USD to EUR on the first screen, and 100 USD to EUR, 50 EUR to GBP and 1 BTC to USD in the currency examples). All gave results without errors, with the same result in the preview and when committed.

Memory checks and random requests

Test	Contents	Result
Function tests (memory-check build)	E01 to E24 (26 items), N01 to N11 and M01 to M19 run with the engine built with memory checks	N and M all passed. E 25/26, with 0 reports from the checks. E22 failed because the process had not ended 1 second after the timeout reply (4.2 seconds); tried again with the memory-check build, the same request gave the timeout reply after 4.6 seconds, and the process ended with exit code 3 0.1 seconds later (which meets the condition)
Random requests (memory-check build)	Requests combining, at random, commands (16 kinds, invalid ones included), expressions (broken, huge and deeply nested expressions, the templates of calculus and matrices, trace() and others, about 80 kinds), values of the wrong type and broken JSON (cut-off lines, extra characters, random bytes, deep nesting, invalid characters). 3,000 each with 11 random seeds, 33,000 in all	Every request answered. The engine never crashed, and the checks reported nothing (0 bytes on standard error)
Random requests (the released engine, qemu)	Requests made the same way, 1,500 each with 4 random seeds, 6,000 in all	Every request answered. The engine restarted twice after timeouts (exit code 3, as designed), and ended in no other way

High precision (all 16 passed; version 1.2.0)

With the precision at its limit of 200000 digits, the released engine (arm64) was run in simulation on a PC. All the digits of the results were checked against values rounded to the same number of digits by another library (Python's mpmath 1.4.1), or against the official qalc built for a PC. The times are those of the simulation; run directly on a PC, they are about one tenth.

ID	Contents	Result
H01	π, e, √2 and ∛2 with 200000 digits: 200,001 characters each, every digit the same as mpmath (1.5 to 3.8 seconds in the simulation)	Pass
H02	sin(1), ln(2) and cos(π/7) with 200000 digits: every digit the same as mpmath (7.7 to 9.5 seconds in the simulation; sin(1) takes 1.2 seconds in qalc on a PC)	Pass
H03	atan(1), exp(1), 1/7 and √(−2) + π with 200000 digits: the same as qalc on a PC (the complex number has 400,006 characters)	Pass
H04	With a precision of 100000 digits too, π, e, √2, sin(1), ln(2), ∛2 and cos(π/7) are the same as mpmath in every digit; atan(1), exp(1) and π × 10^−5 are the same as qalc	Pass
H05	10000! with all its 35,660 digits (0.02 seconds)	Pass
H06	Asking for a precision of 300000 calculates with 200000 digits (kept within the limit)	Pass
H07	Precisions of 1000 and 10 work as before (π has 1,001 and 11 characters)	Pass
H08	π with 200000 digits and digit grouping (266,667 characters, about 400,000 bytes) is not cut	Pass
H09	A result over 450,000 bytes ([π, e; √2, √3]) is cut at the end with "…", truncated:true	Pass
H10	The preview while typing (max_chars 6000) is cut at 6,000 bytes with "…"	Pass
H11	With the precision still at 200000, a 2D graph is calculated with 20 digits in 0.07 seconds (a 3D graph at a precision of 100000 in 0.11 seconds); π afterwards has all its digits (the precision is restored)	Pass
H12	The preview of ζ(3) (1.5 seconds): the watchdog replies with a timeout after 4.6 seconds and restarts the engine	Pass

ID	Contents	Result
H13	An ans of 1000 digits given back to a new engine keeps its 1,001 characters (1.1.0 rounded it to about 38 digits)	Pass
H14	An ans of 200000 digits (the value of π has 400,076 characters) given back to a new engine: 1.2 seconds in the simulation (12.5 seconds before reading with GMP). ans and ans × 2 are the same as before. The same for ln(2) and a complex number (800,155 characters, 2.6 seconds)	Pass
H15	Long numbers read with GMP give the same results as when libqalculate reads them: for 16 values (positive, negative, exponents ±, complex numbers, matrices, values with units, fractions, 200000 digits and so on), ans, ans × 3 − 1 and the value saved again are all the same. π with 200000 digits takes 0.18 seconds (3.40 seconds with libqalculate alone; run directly on a PC)	Pass
H16	Other settings with a precision of 100000 digits: √2 in exact mode stays a symbol, 1/3 in decimal has 100,002 characters, π to hex 83,051 characters and 100000! 100,008 characters	Pass

Changes in 1.3.0: H09 and H10 describe version 1.2.0. In 1.3.0, a result over 450,000 bytes keeps its beginning and its end and leaves out the middle (tail_chars; “The middle of this long result is not shown (N digits)”), and the preview shows up to 450,000 bytes too (detailed design 3.5, 3.8 and 3.9). Q02 checks this.

6.3 Tests of the Android part (Java) and results

Tests with the real engine (all 21 passed; run again with the engine of 1.4.0; Engine.java and StateStore.java have not changed since 1.3.0)

Engine starts the engine running in qemu and uses it.

ID	Contents	Result
J01	init succeeds (about 0.23 seconds)	Pass
J02	Committing 1/3 gives 0.3333333333	Pass

ID	Contents	Result
J03	ans × 3 = 1 in the same engine	Pass
J04	The value of the ans event (1/3) is saved in SharedPreferences	Pass
J05	After the engine is stopped, the next request starts it again, and ans × 6 = 2 (ans is back)	Pass
J06	Stopping a calculation: cancel 1.5 seconds into fibonacci(10^9) gives code=cancelled (1.505 seconds)	Pass
J07	2+2 = 4 after the cancel	Pass
J08	A null request gives code=bad_request	Pass
J09	A request with a line break gives code=bad_request	Pass
J10	A request that is not JSON gives code=bad_request	Pass
J11	The result of rand() equals ans after the engine restarts	Pass
J12	StateStore: content waiting to be written can be read	Pass
J13	StateStore: state/qalc.history.json is made	Pass
J14	StateStore: a name such as “../evil” is refused	Pass
J15	init succeeds with the engine of 1.4.0	Pass
J16	Committing π with a precision of 200000 digits gives 200,001 characters	Pass
J17	Its ans value (interval(lower, upper), 400,076 characters) is saved in SharedPreferences (up to 500,000 characters)	Pass
J18	After the engine is stopped, the next request starts it again and ans comes back with all its digits (200,001 characters, as π; about 2.5 seconds to start, restore and calculate in the simulation on a PC)	Pass
J19	√2 with 1000 digits also keeps its 1,001 characters after a restart (1.1.0 rounded it to about 38 digits)	Pass

ID	Contents	Result
J20	A result over 450,000 bytes ([π, e; √2, √3]) is cut, ending in "…"	Pass
J21	An ans value over 500,000 characters (the same matrix, about 800,000 characters) is not saved (null)	Pass

Tests with a fake engine (all 10 passed; run again with 1.4.0)

A small program that behaves in ways the real engine rarely does (no reply, ending midway, printing broken lines) is used in place of the engine.

ID	Contents	Result
F01	init succeeds	Pass
F02	An engine that does not reply gives code=timeout after the waiting time (timeout 500 + 10 seconds; 10.501 seconds)	Pass
F03	The next request starts a new engine, which replies	Pass
F04	Unrelated lines and lines cut off midway are skipped	Pass
F05	An engine that ends after a request gives code=crash (the request is not sent again)	Pass
F06	The next request succeeds	Pass
F07	A reply with restart=true is passed on as it is	Pass
F08	The next request succeeds with a new engine	Pass
F09	An ans event is passed to the page and its value saved	Pass
F10	An event line is not mistaken for a reply	Pass

MainActivity (the WebView, the Back button, recovering when the screen stops) and the downloads of RatesUpdater need Android, so they are not part of these tests (6.6).

The Java added in 1.3.0 (Pro.java, saveFile and tr in MainActivity, PlayBilling.java) also needs Android and is not part of these tests. For PlayBilling.java, only compiling it against stand-in classes that imitate the types of Google Play Billing Library 8 was checked

(building a version with billing and buying are in 6.6). How the page behaves in each state of Pro was checked in the tests of the page (Q04 to Q18), with a simulation in place of Java.

6.4 Tests of the page and results

The page was opened in Chromium, and test scripts pressed keys and did the rest of the operation to check the results (calculations by the real engine). The tests W, V, U, G, R and L run on the Japanese screen, so the texts they check are the Japanese ones (an English gloss is given in brackets).

Tabs of keys, calculus and matrices on the page (all 26 passed; run again on the Japanese screen of 1.4.0)

The page changed in 1.1.0 (tabs of keys, templates, matrix tables, examples) was checked at several screen sizes.

ID	What	Actual result	Result
W01	Keys at start	4 tabs (“123” chosen), 25 keys, 8 keys in the top row (x , a=b → ans ⌨ ◀▶)	Pass
W02	The first screen is seen whole at 360 × 740	top of the description at 14 px, 5 px left at the bottom, no scrolling; 4 examples	Pass
W03	The tabs switch the keys	Functions (sin cos tan …), Calculus (d/dx d²/dx² ∂/∂y …), Matrix ([ ] ; …), 123 (( ) % …). 25 keys each, and the chosen tab changes color	Pass
W04	The $\int_a^b$ template	integrate(, 0, 1) is typed with the cursor right after “(”. Typing x^2 gives integrate(x^2, 0, 1) = 1/3 ≈ 0.3333333333	Pass
W05	Cursor positions of templates, and ⌫	$\int_a^b$, d²/dx², ∂/∂y, lim, Σ, Π, d/dx, the extremum key and the others all put the cursor at the first argument. An empty template goes with one ⌫	Pass
W06	d²/dx² and ∂/∂y	diff(x^5, x, 2) = 20x³, diff(x^2 × y^3, y) = 3x²y²	Pass

ID	What	Actual result	Result
W07	lim, Σ, Π	limit(sin(x)/x, 0) = 1 with the warning (limit() は完全ではなく、結果が正しくない場合があります), sum(x^2, 1, 10) = 385, product(x, 1, 5) = 120	Pass
W08	The dsolve template, the a=b key, the ∞ key	dsolve(diff(y, x) =) with the cursor before the last “)”. dsolve(diff(y, x) = y) = e^x × C …, solve(x^2 = 16) = [4 −4] with a=b, limit((1 + 1/x)^x, ∞) = e ≈ 2.718281828	Pass
W09	A selection is wrapped in the template	selecting x^2 × sin(x) and pressing $\int_a^b$ gives integrate(x^2 × sin(x), 0, 1); selecting −2y and pressing dsolve gives dsolve(diff(y, x) = −2y)	Pass
W10	“more…”	On Calculus, the function list opens at the calculus category (18 items); on Matrix, at Matrices & Vectors (51 items, with trace). “Insert” types limit() and closes the list	Pass
W11	[1, 2; 3, 4] and A^{-1} with the Matrix keys	The expression is [1, 2; 3, 4]^(−1). The preview shows two 2-column tables (−2, 1, 3/2, −1/2 and −2, 1, 1.5, −0.5). The name read aloud is [−2 1; 3/2 −1/2]	Pass
W12	Matrices in the history, and tapping a result	Two tables in the history. Tapping the result inserts [−2 1; 1.5 −0.5] as text, which gives the same matrix	Pass
W13	det, A^T, I_n	det([1, 2; 3, 4]) = −2, transpose([1, 2, 3; 4, 5, 6]) as a table of 3 rows and 2 columns, identity(3)	Pass
W14	A matrix product; a wide matrix	Product [19 22; 43 50]. The approximate inverse of a 3 × 3 matrix (9 elements) fits the screen with its text shrunk to 0.738em	Pass
W15	A row vector stays text, a column vector becomes a vertical table	[1, 2, 3] + 2 × [4, 5, 6] = [9 12 15] is not a table. The solution of a system, [4/5; 7/5], is shown vertically	Pass

ID	What	Actual result	Result
W16	trace() and searching in Japanese	trace([1, 2; 3, 4]) = 5. Searching the function list for トレース (trace) finds 1 item (trace(Matrix), Matrices & Vectors)	Pass
W17	The warning after an integral with an infinite bound	1/0 after the integral shows ゼロで割っています (Division by zero). The same for an expression with the integral and 1/0 (= 1 + 1 / 0)	Pass
W18	Groups of examples	12 groups (including Derivatives, Integrals, Limits, series and differential equations, Matrices and Vectors)	Pass
W19	⌨	hides the tabs and keys; pressing it again brings them back on the same tab (Matrix)	Pass
W20	Script errors in portrait	0	Pass
W21	Dark theme and the text size Extra large	the keys stay on screen (8 px left at the bottom), and matrices fit where they are shown	Pass
W22	Script errors in the dark theme	0	Pass
W23	Landscape 740 × 360	the tabs (26 px high) and 5 rows of keys (42 px high) fit	Pass
W24	Script errors in landscape	0	Pass
W25	320 × 568	the keys stay on screen; the first screen is shown from its top (scroll position 0)	Pass
W26	Graph of x^3 − 3x and its derivative	both lines drawn; the point tapped on the previous graph disappears	Pass

Examples, Undo, text size, search (all 22 passed; run again on the Japanese screen of 1.4.0)

ID	What	Actual result	Result
V01	Examples on the first screen	60 mph to km/h, 100 USD to JPY, solve(x^2 − 5x + 6 = 0), diff(x^3) (the English screen has 100 USD to EUR)	Pass
V02	Tapping an example	it goes into the input field, and the preview shows = 96.56064 km/h	Pass
V03	The examples panel	titled 式の例・使い方 (Examples and tips); 12 groups, 69 examples, 12 tips (1.4.0 added one about the suggestions)	Pass
V04	An example replaces the expression	30 坪 to m² is typed, the panel closes, and “Undo” is offered	Pass
V05	Undo	brings back 60 mph to km/h, typed before	Pass
V06	Undo after AC	the input field is emptied, and “Undo” brings it back	Pass
V07	Typing after AC	the “Undo” offer goes	Pass
V08	Tapping an expression in the history	the expression it replaced (1+2) can be brought back	Pass
V09	Deleting a line of the history	deleting 3×4 shows the notice “1 行を削除しました” (Line deleted) with an “Undo” button, which puts it back in its place	Pass
V10	Clearing the history	clearing shows the first screen (4 examples); “Undo” brings back 1+2, 3×4 and 5×5	Pass
V11	Text size Extra large	results 22 → 30.36 px, input 26 → 35.88 px (1.38 times), keys stay at 17 px. 20 settings (1.4.0), and the setting is saved	Pass
V12	The text size after a restart	still Extra large (30.36 px); can be set back to Normal	Pass

ID	What	Actual result	Result
V13	Searching functions in Japanese	421 in all. 積分 (integral) puts Integrate first; 最小 (minimum) and min put Min near the top	Pass
V14	Japanese Units in the unit list	10 units (貫, 間, 合, 尺, 升, 寸, 坪, 反, 匁, 里). km and 坪 can be found by their Japanese names	Pass
V15	The “Convert” button of a unit	types “ to 坪”	Pass
V15b	Showing and inserting a compound unit	shown as km/h; the inserted 90km/h is read as 90 kilometers/hour	Pass
V16	Searching constants in Japanese; a variable named constructor	光速 (speed of light) finds c (Speed of Light in Vacuum). A user variable constructor = 5 is listed normally	Pass
V17	The menu	13 items (1.3.0 added QCalc Pro, Language / 言語 and exporting the history), with 式の例・使い方 (Examples and tips) first. The examples can also be opened from About	Pass
V18	“Clear history” with an empty history	only says 履歴はありません (The history is empty), without asking	Pass
V19	Announcements	messages are role=status; “Undo” is a button	Pass
V20	The first screen in the dark theme and landscape	no horizontal overflow (854 px wide)	Pass
V21	Page errors	0	Pass

Reusing the history and converting input (all 12 passed; run again on the Japanese screen of 1.4.0)

ID	What	Actual result	Result
U01	Tapping the result of 7/3	inserts (7/3)	Pass
U02	Tapping the result of 255 to hex (0xFF)	inserts 255	Pass

ID	What	Actual result	Result
U03	Tapping the result of x^2 = 4	inserts [2 −2]	Pass
U04	Tapping the result of 2x + 3 = 7	inserts 2	Pass
U05	Tapping the result of 5 ft to m	inserts (1.524 m)	Pass
U06	Tapping the result of sqrt(2)	inserts 1.414213562	Pass
U07	Tapping the result of 1/0.25	inserts 4	Pass
U08	Inserting the result of 7/3 after “2×” and committing	= 14/3 = 4 + 2/3 ≈ 4.666666667	Pass
U09	Typing the full-width １０＋５	converted to 10+5	Pass
U10	Typing 100円	converted to 100¥	Pass
U11	Committing 100¥	= ¥100	Pass
U12	While typing q = 5	shows “＝ を押すと実行します（定義など）” (press = to run it: definitions and the like) and does not run it	Pass

Operation and display (all 9 passed; run again on the Japanese screen of 1.4.0)

ID	What	Actual result	Result
U13	A plain click on a key (screen readers, keyboard operation)	typed once	Pass
U14	A click following a touch	typed once	Pass
U15	The keys stay on screen with a preview of 2,600 characters	bottom of the = key at 846 / screen 854	Pass
U16	Graph of x + 1e17	drawn without freezing	Pass
U17	Zooming a graph 60 times	does not freeze	Pass
U18	Searching the unit list for 円 (yen)	finds Japanese Yen	Pass
U19	Searching the function list for command	command(), which cannot be used, is not listed (0 items)	Pass
U20	Characters typed while calculating	stay after committing (123)	Pass
U21	The message for 1/0	ゼロで割っています (Division by zero)	Pass

3D graphs (all 10 passed; run again with 1.4.0)

The graph panel was switched to 3D and operated, and the drawings were also checked as pictures (screen size 405 × 720).

ID	What	Actual result	Result
G01	Switching to 3D and drawing sin(x) cos(y) (−5 to 5, Normal 41 × 41)	the surface, the axes and the color bar (z: −1.00 to 1.00); no message	Pass
G02	Turning by dragging (across, up and down)	the direction changes, and the axes and ticks turn with it	Pass
G03	Tapping the surface	“sin(x) cos(y)：x = 1.5, y = -1.25, z = 0.3145324757” and a red point	Pass
G04	A surface with a pole, 1/(x^2 + y^2) (−2 to 2)	values too large near the pole are left out of the color range (z: 0.1 to 23.0)	Pass
G05	Ripples sin(sqrt(x^2 + y^2)) / sqrt(x^2 + y^2) (−10 to 10, Fine 61 × 61)	the central peak and the rings are drawn	Pass
G06	Two surfaces, x^2/10 − y^2/10 and 2 (Coarse 25 × 25)	surfaces in 2 colors, with a legend	Pass
G07	sqrt(x) (not real for x < 0)	only the part with x ⩾ 0 is drawn (z: 0 to 2.24)	Pass
G08	A wrong y range	“y の範囲を確認してください” (Check the range of y)	Pass
G09	Back to 2D, closing the graph and opening it again	opens in 2D (the last mode is remembered)	Pass
G10	Dark theme and landscape	shown without breaking; 0 console errors or warnings	Pass

Fixes from the review (all 10 passed; run again with 1.4.0)

Checked that the points fixed in the code review stay fixed.

ID	What	Actual result	Result
R01	◀ at the end of “a*x*b”	the cursor goes 4 → 3	Pass

ID	What	Actual result	Result
R02	◀ again (x is a surrogate pair, 2 characters)	3 → 1 in one press (it does not stop inside the character)	Pass
R03	▶	1 → 3 in one press	Pass
R04	◀ with a selection	the cursor goes to the start of the selection (1)	Pass
R05	▶ with a selection	the cursor goes to the end of the selection (3)	Pass
R06	1/(x−x) in a 2D graph (division by zero)	the English message (Division by zero) is not shown raw on the Japanese screen	Pass
R07	sqrt(−1 − x^2 − y^2) in 3D, with no real values	“(値なし)” (no values) in the legend, no color bar, and “実数の値がありません（x と y の範囲を確認してください)” (There are no real values)	Pass
R08	Names for screen readers on the graph canvases	2D グラフ and 3D グラフ (2D graph, 3D graph)	Pass
R09	Turning 2 surfaces of 61 × 61 for 60 frames	20 ms per frame on this PC (21 ms in 1.2.0), turning without freezing	Pass
R10	Page errors	0	Pass

Showing and saving long results (all 32 passed; version 1.2.0; run again on the Japanese screen of 1.4.0)

With the precision at 200000 digits, the page was run in Chromium (a 412 × 915 screen). The calculation engine was tried both as released (arm64, simulated on a PC) and as the memory-check build for a PC, and all 32 passed with both. The tests of reusing the history (12) and of operation and display (9) were also run again on the page of 1.2.0, and all passed.

ID	Contents	Result
L01 to L02	The description of the setting Precision says 2 to 200000 digits, and the field allows up to 200000	Pass
L03 to L06	300000 becomes 200000, and 1 becomes 2; 1000 and 200000 stay. All are saved	Pass
L07	“200000” fits in the field	Pass

ID	Contents	Result
L08 to L09	Committing π shows all its digits (200,003 characters, with “≈ ”), with the right end	Pass
L10 to L11	A long result scrolls inside its frame, which is at most 70 % of the history's height (275 / 393 px)	Pass
L12	The keys are not pushed off the screen	Pass
L13	The text read aloud is only the beginning (204 characters)	Pass
L14	The heading of the long-press menu is only the beginning (304 characters)	Pass
L15	“Copy result” puts all 200,001 characters on the clipboard	Pass
L16	Tapping a result of 200000 digits does not insert it, and says to use the ans key	Pass
L17	While typing, ans × 2 shows the first 6,000 characters and “…”	Pass
L18	Committing ans × 2 shows all digits	Pass
L19	After reloading the page, both results come back with all their digits	Pass
L20 to L25	With 12 results of 200000 digits, the newest 5 keep all their digits and the other 7 the first 2,000 characters and “…”. 1,015,339 characters are saved, and the page shows the same	Pass
L26 to L27	Short results (1 + 2 = 3) are as before, with no scrolling in the frame	Pass
L28	The complex number √(−2) + π (real and imaginary parts of 200000 digits, 400,008 characters) is shown whole	Pass
L29 to L30	π with digit grouping (266,669 characters) is shown whole, and all 266,667 characters can be copied	Pass
L31	Setting the precision back to 10 gives π ≈ 3.141592654	Pass
L32	No script errors	Pass

Changes in 1.3.0: L17 (the first 6,000 characters of ans × 2 while typing) is replaced by Q02, as the preview now shows all the digits too, and L29 to L30 (π with digit grouping) by Q03, as the decimals are now grouped with narrow spaces instead of commas. In 1.4.0 the 32 items were run again on the Japanese screen, and all passed. Two of them are now checked against the present behavior: the heading of L14 begins with the date and time since 1.4.0 (the date and time, the expression and the beginning of the result, 322 characters; T19), and L17 checks, like Q02, that all the digits (200,003 characters) are shown while typing.

The page of 1.3.0 (all 27 passed; run again with 1.4.0)

The features added and the behavior changed in 1.3.0 were checked in Chromium (412 × 915). Calculations use an engine built for a PC from the same source; buying QCalc Pro uses a simulation in place of Java (switching the kind of billing and the state of the purchase). Q01 to Q19 cover QCalc Pro, color themes, exporting, the preview and digit grouping; Q20 to Q26 the language of the screen; Q27 the check of the English screen.

ID	What	Actual result	Result
Q01	First start	English screen. The menu starts with “Examples and tips” and has “QCalc Pro” (test build) and “Export history (CSV)”	Pass
Q02	Preview of 4587!	the preview also shows all 14,806 digits, as the history does (1,144 zeros at the end)	Pass
Q03	Digit grouping	1,234,567; 12.345 km (no comma before the unit); 3.141 592 654 (no commas in the decimals)	Pass
Q04	The Pro panel of the test build	“QCalc Pro is unlocked”, 5 features, no PRO marks	Pass
Q05	Color theme Green	data-theme green, accent color #15803d, saved	Pass
Q06	Exporting the history	UTF-8 with a byte order mark, the headings and 4 rows; numbers without grouping; no cell that would run as a formula	Pass
Q07	Without Pro: the PRO mark	a PRO mark on “Export history”	Pass
Q08	Without Pro: color themes	“Color themes are a QCalc Pro feature”, and the theme stays Blue	Pass

ID	What	Actual result	Result
Q09	Without Pro: precision 5000	set back to 1000, with “A precision above 1000 digits is a QCalc Pro feature”	Pass
Q10	Without Pro: the Pro panel	“Buy QCalc Pro (¥300)” with the price, and “Restore a purchase”	Pass
Q11	Restoring without a purchase	“No purchase of QCalc Pro was found”	Pass
Q12	After buying	“QCalc Pro is unlocked. Thank you!”; the Pro panel shows it unlocked; the PRO marks go	Pass
Q13	3D after buying	the 3D graph opens	Pass
Q14	Without Pro: 3D	“3D graphs are a QCalc Pro feature”, and the 2D graph stays	Pass
Q15	Without Pro: exporting	“Exporting the history is a QCalc Pro feature”	Pass
Q16	A version without billing	no “QCalc Pro” item, and the color themes (Purple) can be used	Pass
Q17	The Pro panel in Japanese	QCalc Pro を購入（¥300）; the first feature is 高精度 (High precision) and the fourth 履歴の書き出し (Export the history)	Pass
Q18	Dark theme and Orange (bought, Japanese)	accent color #fb923c	Pass
Q19	Page errors (Q01 to Q18)	0	Pass
Q20	First start	English; 1 history item, and 7*6 being typed	Pass
Q21	日本語 (Japanese) in the menu	the screen becomes Japanese; the history and 7*6 stay	Pass
Q22	English in the settings (the first setting)	back to English; the history and 7*6 stay	Pass
Q23	Page errors (Q20 to Q26)	0	Pass
Q24	A test build left lang "ja" in the settings, with no qalc.lang	the app starts in English (the other settings stay)	Pass

ID	What	Actual result	Result
Q25	Japanese chosen in the menu, then the app opened again	still Japanese (qalc.lang is "ja")	Pass
Q26	English chosen again	stays English (qalc.lang is "en")	Pass
Q27	No Japanese left on the English screen	the first screen, the 4 tabs of keys, the menu, 10 panels (with 3D and the licenses), and the results and history of 5 calculations were scanned: the only Japanese is “Language / 言語”, “日本語” and the names of the Japanese units (坪 and the like)	Pass

The page of 1.4.0 (all 27 passed; version 1.4.0)

The features added and the behavior fixed in 1.4.0 were checked in Chromium (412 × 915; T24 in landscape, 780 × 360). Calculations use an engine built for a PC from the same source. T01 to T24 and T27 use the English screen, T25 and T26 the Japanese screen. Typing with the phone keyboard is simulated by sending input events to the input field.

ID	What	Actual result	Result
T01	Typing “sin” with the keyboard	sin(), sinh() and sinc() come first	Pass
T02	Tapping a function	the name and brackets are written, with the cursor inside the brackets; the suggestions go	Pass
T03	Typing “integ” in the middle of an expression and tapping the suggestion	only the letters typed are replaced by integrate() (2 + integrate())	Pass
T04	A single letter; a variable or unit written in full (pi, kg)	nothing is offered	Pass
T05	sin typed with the on-screen keys	no names are offered	Pass
T06	Typing “pound”	lb and GBP are found by a word of their full names	Pass
T07	Moving the cursor away from the name (◀)	the names go	Pass

ID	What	Actual result	Result
T08	→ after “5 km”	units of the same kind, common ones first (m, cm, mm, mi, ft); km is not offered	Pass
T09	Tapping the target mi	the field reads “5 km to mi” and the preview shows the converted result; the suggestions go	Pass
T10	→ after “100 USD” (English screen)	EUR, JPY, GBP in this order; USD is not offered	Pass
T11	→ after “255”, which has no unit	fraction, hex, bin, oct in this order; tapping hex gives “255 to hex” and 0xFF	Pass
T12	°C, km/h, kPa (with a prefix) and hours (a unit by its name)	°F and K; m/s, mph and knot; Pa, bar and atm; s, min and d come first	Pass
T13	Units of different kinds (5 km / 2 h); a function (sin(30°))	nothing for the former; for the latter, sin is not read as units, and the ways to write a number are offered	Pass
T14	Typing letters after “to”	“5 km to m” offers only the same kind (mm, mi and others), and “255 to he” offers hex. After a unit written in full (30 tsubo to m²), units of other kinds (such as m²·A) are not offered	Pass
T15	→ after the Japanese unit “30 坪”	the area units m², km² and ha come first	Pass
T16	An expression ending with “to” (255 to, 5 km to)	the preview shows the result of what is before “to” (= 255, = 5 km)	Pass
T17	= with “7 km to”	“7 km” is calculated and added to the history	Pass
T18	Calculating several times	only the new line is added to the history on the screen, and the earlier lines are not built again; a matrix is shown as a table	Pass
T19	Long-pressing a history line	the heading of the menu begins with the date and time, followed by the expression and the result	Pass

ID	What	Actual result	Result
T20	Deleting a line, then Undo	the chosen line goes and comes back in the same place	Pass
T21	Calculating with 300 lines in the history	the oldest line goes and the new result is shown (still 300 lines)	Pass
T22	The setting “Light or dark”	Dark and Light give those colors whatever the phone’s setting; “Same as the phone” follows the phone	Pass
T23	Choosing Light on a phone set to dark, then opening the app again	stays light	Pass
T24	Typing a long expression in landscape	the keys and the buttons at the top stay on the screen	Pass
T25	The Japanese screen	積分 finds integrate(); JPY is the first currency offered; for lengths, the Japanese units (寸 and the others) come after the common units	Pass
T26	Suggestions on the Japanese screen	the descriptions are the Japanese names; the name for screen readers is 候補	Pass
T27	π with a precision of 300, and a polynomial of 14 terms in x, y and z	the long number begins on the line of its “≈” and is cut at the end of each line; the result without a long number (the polynomial) wraps between words	Pass

No script errors occurred on the page during T01 to T27.

6.5 Checks of the APK and App Bundle and results

Files checked: the test APK “out/QCalc-1.4.0-debug.apk” (3,340,853 bytes, SHA-256 b75277a6…9717) and the App Bundle for Google Play “out/QCalc-1.4.0.aab” (3,333,494 bytes, SHA-256 b03c9039…7695). The engine inside is 7,588,488 bytes (SHA-256 ac8f1bd9…b9a7). K01 to K10 are mainly about the test APK and the engine, K11 to K22 mainly about the App Bundle and the device APKs made from it with bundletool. bundletool was downloaded, so it was run in an environment cut off from the network. All 22 items were run again with the two files of 1.4.0 (2026-11-13).

ID	Contents	Actual result	Result
K01	Signature of the test APK (apksigner verify)	verified with v1, v2 and v3 (including the checks from Android 6 on)	Pass
K02	Alignment (zipalign -c -p 4)	verified	Pass
K03	Package information (aapt2 dump badging)	io.github.hiroshitsu.qcalc.debug, versionCode 6, versionName 1.4.0-debug, minSdk 24, targetSdk 36, activity started io.github.hiroshitsu.qcalc.MainActivity, arm64-v8a	Pass
K04	Permissions	only INTERNET (exchange rates) and VIBRATE (key vibration)	Pass
K05	Contents up to date	the engine and the 4 files of the page (app.js, index.html, licenses.js, style.css) are the same as the sources	Pass
K06	Nothing extra	29 entries besides the signature files (1.4.0 added the 5 background layers of the icon)	Pass
K07	How the engine is stored	compressed (with extractNativeLibs, Android takes it out in a form that can be run when installing)	Pass
K08	Form of the engine	an AArch64 executable (ELF, EXEC)	Pass
K09	16 KB pages	4 LOAD segments, all aligned to 0x4000	Pass
K10	Statically linked	needs no shared library	Pass
K11	Signature of the App Bundle (jarsigner -verify)	verified	Pass
K12	Signing key	the upload key (CN=QCalc upload key, O=HiroshiTsu), not the test key	Pass
K13	The same contents in the App Bundle and the APK	the engine, the page files and classes.dex are the same	Pass
K14	Layout of the App Bundle	only BundleConfig.pb and base/ (manifest, dex, lib/arm64-v8a, assets, res, resources.pb)	Pass
K15	No key or password file in the App Bundle	none	Pass

ID	Contents	Actual result	Result
K16	No key or password file in the APK	none	Pass
K17	bundletool validate	accepted	Pass
K18	Manifest of the App Bundle	io.github.hiroshitsu.qcalc, versionCode 6, versionName 1.4.0, minSdk 24, targetSdk 36	Pass
K19	Manifest of the App Bundle (continued)	extractNativeLibs; only the INTERNET and VIBRATE permissions; not debuggable	Pass
K20	bundletool build-apks	APKs can be made for an arm64 Android 16 phone	Pass
K21	The engine in the APK delivered to a phone	base-master.apk has the same engine	Pass
K22	Information of the APK delivered to a phone	io.github.hiroshitsu.qcalc, versionCode 6, versionName 1.4.0, label QCalc, activity started io.github.hiroshitsu.qcalc.MainActivity	Pass

6.6 Tests not yet run, and known limits

Tests not yet run (they need a phone or an Android development environment)

Item	Reason, and what to check
Installing and starting on a phone (acceptance criterion A-02)	Needs checking on the AQUOS sense8 (Android 16), including that the engine starts from nativeLibraryDir. Check both the test APK and the version installed from Google Play through internal testing
Use with TalkBack (A-10)	What is read aloud, and operating the keys, the history and the menu
Real exchange-rate downloads (A-11)	The test environment cannot reach outside servers, so downloading from the 3 sources and checking the data have not been tried
Typing with the phone's Japanese keyboard	The automated tests simulate input events. Check characters being converted, Enter, and what happens when the keyboard closes

Item	Reason, and what to check
Display in Android System WebView	Differences from Chromium on a PC; broken text or layout
The Calculus and Matrix keys and matrix tables on a phone	Superscripts and subscripts ($\int_a^b$, A^T, I_n) and table lines in Android System WebView. TalkBack reads a matrix as one line of text ([1 2; 3 4]) by design (detailed design 3.9), so check how that sounds too
Turning the screen, switching dark mode, Back (with predictive back), vibration, copying	MainActivity has not run on a PC. Also check that on Android 15 and later the page does not overlap the status bar, the navigation bar or the keyboard
Restoring after the app is closed	The history, the expression being typed, the settings and ans stay (saving itself is checked in 6.3)
Low memory, long use	Including re-creating the screen after it stops
Each Android version from 7.0 to 15	Phones near the lowest version supported
Building and running in Android Studio (Gradle)	Not checked, as Gradle and the Android SDK cannot be downloaded in the test environment. Check syncing, running the debug build, signing the release and installing it over the previous one
Phones with 16 KB pages	The engine is aligned to 16 KB (K09) and meets Google Play's rule, but has not run on a phone with 16 KB pages
Speed of 3D graphs on a phone	On the PC, turning 2 surfaces of 61 × 61 takes 21 ms per frame (R09). Turning may be slower on some phones; the Coarse grid (25 × 25) is lighter
Speed of high precision and copying on a phone (checking A-19 on a phone)	On the PC, sin(1) with 200,000 digits takes 1.2 seconds and π 0.4 seconds. Check the time on a phone, copying a result of 200,000 digits (about 270,000 characters with digit grouping) to the clipboard, and restoring ans after the app is closed

Item	Reason, and what to check
Google Play's review and closed testing	Upload the .aab to Play Console and check with the pre-launch report and the testers' phones (the release flow is in 5.7)
Buying QCalc Pro (a version with billing)	Building with Gradle (qcalc.billing=true) has not been checked, as the test environment cannot reach Google's Maven repository. Create the product qcalc_pro in Play Console and, with a license tester's account and a version installed from internal testing, check buying, restoring, pending payments, refunds and the handling of earlier users (legacy)
The language and exporting the history on a phone	The app starts in English, and switching with Language / 言語 keeps the history and the expression being typed; the language of the texts that Java shows itself (the copy notice and the like); saving through the system file picker (to the phone and to places such as Google Drive)
Suggestions and Light or dark on a phone (1.4.0)	Names are offered while typing with the phone keyboard (including Japanese input) and are written when tapped. With the setting "Light or dark", the colors of the status bar and the navigation bar and the brightness of their icons match the page (setAppearance and applyColors in MainActivity have not run on a PC; only compiling was checked)

Known limits

- It runs only on 64-bit ARM (arm64) phones. 32-bit and x86 phones are not supported.
- The screen is in English or Japanese only. Japanese is used only when chosen with "Language / 言語" in the menu or the settings (the phone's language is not followed).
- QCalc Pro can be bought only in versions with billing (Gradle with qcalc.billing=true). The versions made by build-aab.sh have no billing, and all the Pro features can be used (detailed design 3.20).
- libqalculate's messages are in English. On the Japanese screen, only the 28 common ones (57 kinds, counting the texts of Java and the engine) are shown in Japanese (detailed design 3.14). On the English screen, all are shown in English.

- plot(), command(), load() and export() cannot be used. For graphs, use “Graph” in the menu (2D: up to 6 functions of x; 3D: up to 4 functions of x and y).
- Temporary variables (such as x = 5) and ans2 to ans5 are lost when the app is closed or the engine restarts. ans comes back.
- The precision is at most 200000 digits. The more digits, the longer the calculation; heavy functions such as ζ(3) go over the commit time limit (20 seconds) and are stopped. The preview while typing is stopped after 1.5 seconds, so with a high precision it often shows “…”.
- A result over 450,000 bytes (UTF-8) keeps its beginning and its end and leaves out the middle (“…” and “The middle of this long result is not shown (N digits)”; 1.3.0). Android limits copying to the clipboard to about 1 MB at a time, so a complex number with real and imaginary parts of 200000 digits and the like may not be copied. The history keeps all the digits of the newest results up to 1,000,000 characters in total, and the ans that stays after the app is closed is a value of up to 500,000 characters.
- Exchange rates depend on the source services. When they cannot be fetched, the rates fetched last time (or, if there are none, those built into the engine) are used.
- libqalculate always adds the warning “limit() is incomplete and unreliable.” to limit() (the result is still shown). dsolve() solves first-order differential equations only. Sums with an upper bound of ∞ (infinite series) are not calculated and come back as they are.
- One-letter names (a, b, m, t and others) may be read as units. In calculus, use x, y, z and n as variables.
- No function gives the eigenvalues of a matrix directly; find them from the characteristic equation solve(det(M − x × identity(n)) = 0) (one of the examples). The rank of a matrix is rk() (rank() is a different function that returns ranks).
- Matrices are shown as tables when they have 2 rows or more and up to 400 elements. Larger ones and one-row vectors are shown as the one line of text the engine returns.
- The phone's text size setting is not followed (the app's setting “Text size” has 3 steps).
- The test scripts are not part of the source.